Chapter 13 — Reactive Programming and Spring WebFlux

Most Spring applications spend their time **waiting** — for the database, another HTTP service, a **message broker** (Kafka, RabbitMQ: a server storing messages between producers and consumers). In the classic model (Spring MVC on Tomcat) each request holds one **thread** (an independent execution path with its own call stack) from start to finish, waiting included. **Reactive programming** does the opposite: no thread ever waits; work is a pipeline of steps, and a few threads run each step when its data arrives. **Spring WebFlux** is Spring's web framework on that model, using **Project Reactor**. Since Java 21 a second route to the same **scalability** exists: **virtual threads** (lightweight JVM-managed threads) make waiting cheap instead of avoiding it. This chapter teaches the reactive model, then **measures** blocking MVC on virtual threads against WebFlux.

All code is in `code/`. Spring Boot **4.1.1** (Framework 7.0, Reactor 3.8 from the Reactor 2025.0 release train — library versions released together, Reactor Netty — a non-blocking server on Netty, R2DBC H2 1.1), Java 25:

Project	Shows
`reactor-basics`	Reactor only (no Spring context): creation, operators, errors, virtual time, backpressure, schedulers, sinks, context — all as `StepVerifier` tests
`webflux-app`	a book API: annotated controller, functional routes, SSE, `WebClient`, R2DBC repository + `DatabaseClient` + reactive transaction
`load-mvc`, `load-webflux`	the same I/O-bound endpoint (spends its time waiting on I/O, not computing) under load: MVC on platform threads (ordinary OS threads), MVC on virtual threads, WebFlux

Output blocks are real `mvn -q -B -ntp verify` output (Spring log lines trimmed to `...`). **All timings are machine-dependent** (14-core Apple M4 Pro, load generator in the same JVM as the server).

> **Prerequisites and links.** The mental model is Node's: JavaScript handbook Chapter 12 — Asynchronous JavaScript ("The event loop", "Promises"). A `Mono` is roughly a lazy, cancellable Promise; a `Flux` a lazy async stream. Virtual threads and `CompletableFuture` (Java's Promise): Java handbook Chapter 15. Spring MVC: Chapter 5; blocking `RestClient`: Chapter 6; Spring testing: Chapter 12.

13.1 Why reactive? The cost of waiting

The problem it solves

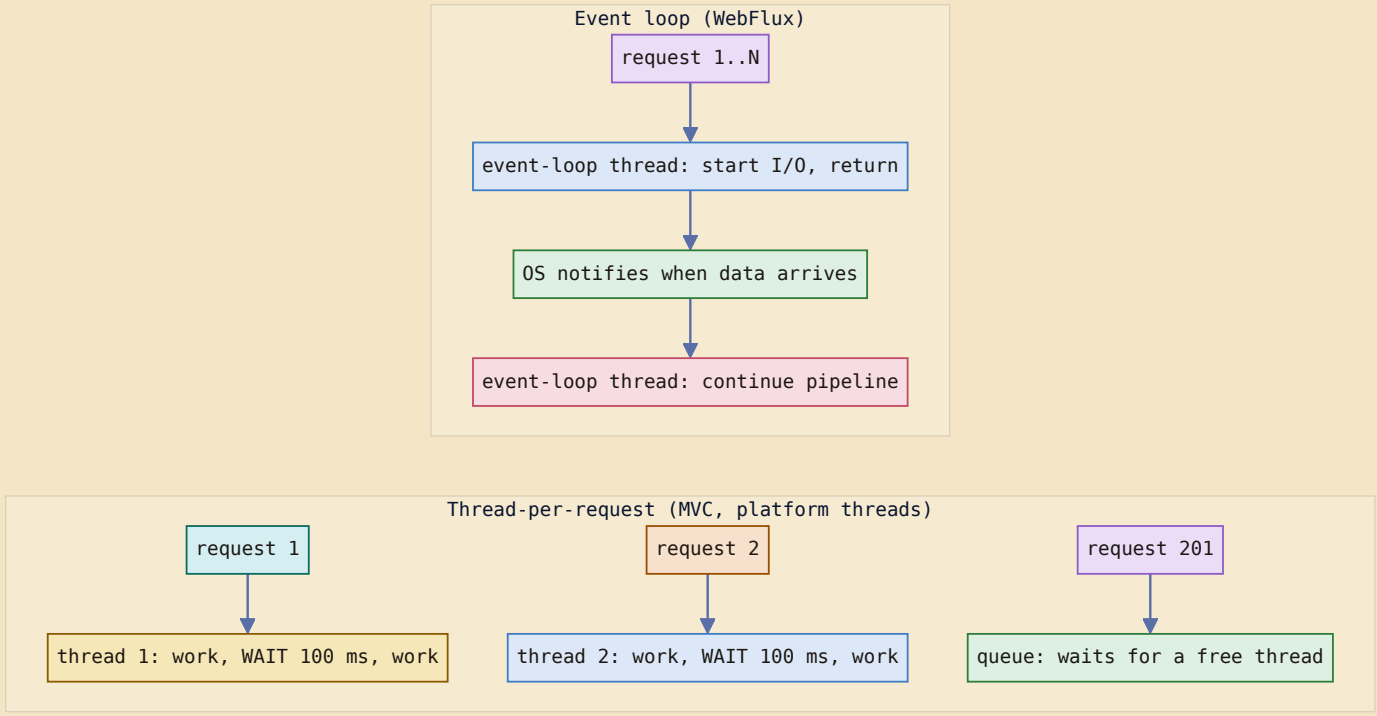
In Spring MVC a request is served by **one thread from start to finish** (*thread-per-request*). If the handler waits 100 ms for the database, the thread is unavailable to anyone else for those 100 ms. Tomcat's worker pool has **200 threads** by default, so at most 200 requests are *in progress* (in flight) at once, and with 100 ms of waiting each the server tops out near **200 / 0.1 s = 2,000 requests per second**, however idle the CPU. More platform threads cost memory (each **stack** reserves up to ~1 MB) and OS scheduling overhead, so "just add threads" stops working at a few thousand.

What reactive does instead

I/O is reading or writing outside the program (network, disk, database). **Blocking** I/O stops the calling thread until data arrives; non-blocking I/O never parks a thread on a **socket** (an open network connection):

1. The handler starts the call and registers **what to do when the answer arrives**.
2. The thread immediately returns to the **event loop** for other work.
3. When the OS reports data ready (epoll/kqueue on Linux/macOS), an event-loop thread runs the next pipeline step.

A handful of event-loop threads (one per core by default in Reactor Netty) keep thousands of requests in progress — like a waiter who hands the order to the kitchen and serves other tables until the bell rings. This is exactly Node.js: the JavaScript event-loop rules (never block the loop; **callbacks** run when I/O completes; one slow **synchronous** function freezes everyone) apply to WebFlux too.



The price

- Code becomes a **pipeline of operators**. Stack traces, debugging and `ThreadLocal`-based libraries (per-thread variables: MDC logging — the Mapped Diagnostic Context adding fields like a request id to log lines — security context, transactions) need reactive-aware replacements.
- **Everything** on the request path must be non-blocking: server, HTTP client, database driver. One blocking call on an event loop stalls every request that loop serves (§13.15 measures this).
- Since Java 21, **virtual threads** remove the thread-count limit for ordinary blocking code — removing the main *performance* reason to go reactive for many apps.

13.2 Reactive Streams: the contract

What it is

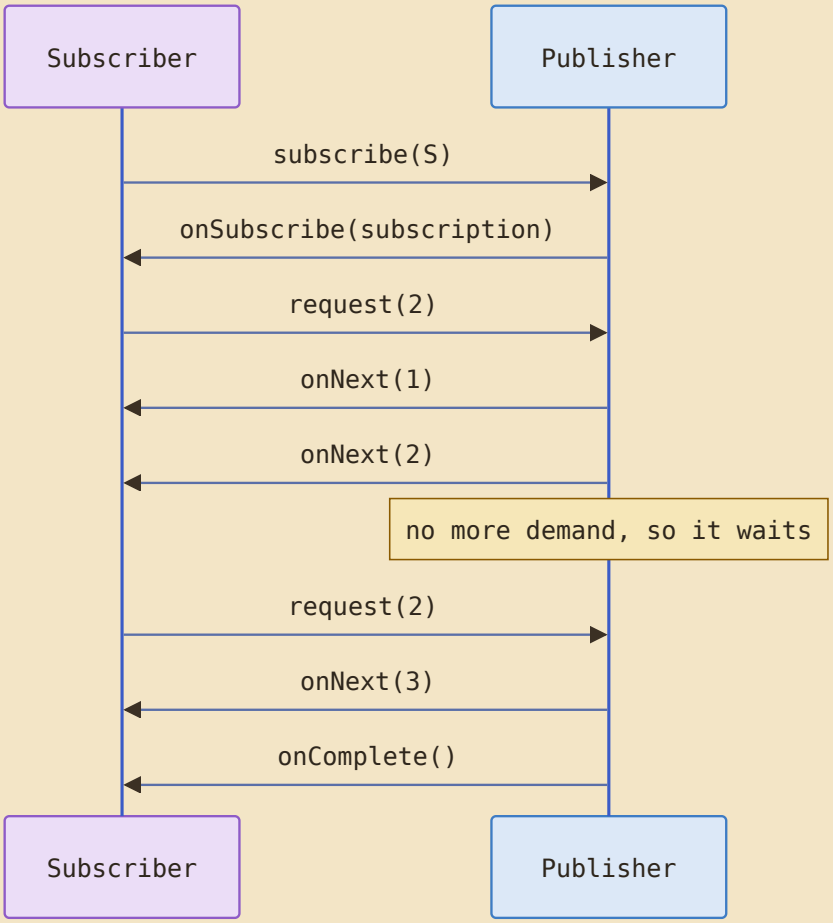
Reactive Streams (1.0, 2015) is a small specification for **asynchronous stream processing with non-blocking backpressure** (*asynchronous*: the caller doesn't wait; the result comes later). Four interfaces, copied into Java 9 as `java.util.concurrent.Flow`:

Interface	Role	Methods
`Publisher<T>`	produces items	`subscribe(Subscriber)`
`Subscriber<T>`	consumes items	`onSubscribe(Subscription)`, `onNext(T)`, `onError(Throwable)`, `onComplete()`
`Subscription`	the link between the two	`request(long n)`, `cancel()`
`Processor<T,R>`	both a subscriber and a publisher	—

Backpressure: the **consumer controls the flow rate** — the publisher sends at most as many items as requested, so a slow consumer is never flooded.

How it works

1. `publisher.subscribe(subscriber)` — nothing flows yet.
2. The publisher calls `subscriber.onSubscribe(subscription)`.
3. The subscriber calls `subscription.request(n)`, meaning "I can take n more".
4. The publisher sends up to n `onNext` signals, then waits for more demand.
5. The stream ends with at most one terminal signal — `onComplete` **or** `onError` (none if cancelled or infinite) — or the subscriber calls `cancel()`.



You rarely implement these yourself: **Project Reactor** (Spring's choice), RxJava, Mutiny, Akka Streams and drivers such as R2DBC do, and interoperate through the shared contract.

13.3 Project Reactor: `Mono` and `Flux`

What it is

Project Reactor implements Reactive Streams with two publisher types and hundreds of operators:

Type	Emits	Like	Examples
`Mono<T>`	0 or 1 item, then complete (or an error)	a lazy `CompletableFuture<Optional<T>>` / a lazy Promise	find by id, save, an HTTP response
`Flux<T>`	0..N items (possibly infinite), then complete (or an error)	a lazy async `Stream<T>` / async iterator	query results, SSE events, Kafka records

An **operator** (`map`, `filter`, `flatMap`, ...) returns a *new* publisher wrapping the previous one; chaining **assembles a pipeline**. Nothing runs until something **subscribes**.

Creating publishers — `CreationAndLazinessTest`

```
StepVerifier.create(Mono.just("one")).expectNext("one").verifyComplete();
StepVerifier.create(Mono.empty()).verifyComplete();                      // 0 items, then complete
StepVerifier.create(Flux.range(1, 3)).expectNext(1, 2, 3).verifyComplete();
StepVerifier.create(Flux.fromIterable(List.of("a", "b"))).expectNext("a", "b").verifyComplete();
StepVerifier.create(Flux.error(new IllegalStateException("boom")))
        .expectErrorMessage("boom").verify();
```

Other factories: `Mono.fromCallable(() -> ...)` (wrap a computation), `Mono.fromFuture(cf)`, `Flux.interval(Duration)` (ticks on the parallel scheduler — a **scheduler** decides which threads run work, §13.8), `Flux.create(sink -> ...)` (bridge callback APIs), `Mono.delay(Duration)`.

Nothing happens until you subscribe

```
@Test
void nothingHappensUntilYouSubscribe() {
    AtomicInteger calls = new AtomicInteger();
    Flux<Integer> pipeline = Flux.range(1, 3)
            .map(i -> { calls.incrementAndGet(); return i * 10; });

    System.out.println("after assembling the pipeline, map ran " + calls.get() + " times");

    List<Integer> out = pipeline.collectList().block();   // block() subscribes (fine in a test)
    System.out.println("after subscribing: " + out + ", map ran " + calls.get() + " times");

    pipeline.blockLast();                                 // cold: a 2nd subscriber re-runs everything
    System.out.println("after a second subscriber, map ran " + calls.get() + " times");
}

@Test
void justIsEagerDeferIsLazy() {
    AtomicInteger dbHits = new AtomicInteger();
    Mono<Integer> eager = Mono.just(dbHits.incrementAndGet());                // runs NOW, once
    Mono<Integer> lazy  = Mono.defer(() -> Mono.just(dbHits.incrementAndGet())); // runs per subscribe
    System.out.println("hits after building both monos: " + dbHits.get());
    eager.block(); eager.block(); lazy.block(); lazy.block();
    System.out.println("hits after 2x eager + 2x lazy subscribe: " + dbHits.get());
}
```

Output:

```
just/empty/range/fromIterable/error verified
after assembling the pipeline, map ran 0 times
after subscribing: [10, 20, 30], map ran 3 times
after a second subscriber, map ran 6 times
hits after building both monos: 1
hits after 2x eager + 2x lazy subscribe: 3
```

- **Assembly vs subscription.** Building the chain does nothing; subscribing runs it. In WebFlux **the framework subscribes** to the `Mono`/`Flux` your controller returns. A pipeline neither returned nor subscribed never runs — the classic bug: a controller calls `repo.save(book);` and returns `Mono.just(book)`, saving nothing. Return `repo.save(book)`.
- **Cold publishers** replay the whole pipeline per subscriber: two subscribers = two database queries.
- **`Mono.just(expr)` evaluates `expr` immediately** at assembly, even if nobody subscribes. Use `Mono.defer(() -> ...)` or `Mono.fromCallable(...)` for per-subscription work.

`block()`

`block()` subscribes and **waits**. Fine in tests, `main` methods and at the edge of a non-reactive app; inside a pipeline or on an event-loop thread it is a bug: Reactor throws

`IllegalStateException("block()/blockFirst()/blockLast() are blocking, which is not supported in thread reactor-http-nio-...")`.

13.4 Operators

The ones you need daily

Goal	Operator	Note
transform each item (sync)	`map`	must not block or return a publisher
keep some items	`filter`, `take(n)`, `skip(n)`, `distinct`	
async call per item	**`flatMap`**	runs inner publishers **concurrently**, output **interleaved**
async call per item, in order, one at a time	**`concatMap`**	sequential; slower
async per item, concurrent, output in source order	**`flatMapSequential`**	buffers early results
combine by position	`zip`, `zipWith`	waits for one from each
one after another	`concat`, `concatWith`	
as they come	`merge`, `mergeWith`	
fold to one value	`reduce`, `count`, `collectList`, `collectMap`	returns `Mono`
fallback if empty	`switchIfEmpty`, `defaultIfEmpty`	common for 404s
group	`buffer(n)`, `window(n)`, `groupBy`	
side effects (logging, metrics)	`doOnNext`, `doOnError`, `doOnSubscribe`, `doFinally`	don't put business logic here
ignore values, wait for completion	`then()`, `then(mono)`	"do A, then B"

`flatMap` vs `concatMap` vs `flatMapSequential` — `OperatorsTest`

Three users; fetching the orders of user N takes (4 − N) seconds per order, so user 3 answers first:

```java
/** Simulated remote call: user N answers after (4 - N) seconds, so later users answer first. */
static Flux<String> ordersOf(int user) {
    return Flux.just("u" + user + "-a", "u" + user + "-b")
            .delayElements(Duration.ofSeconds(4 - user));
}

@Test
void flatMapInterleavesConcatMapKeepsOrder() {
    // withVirtualTime: the delays run on a fake clock, so this 12-"second" test takes milliseconds
    StepVerifier.withVirtualTime(() -> Flux.range(1, 3).flatMap(OperatorsTest::ordersOf).collectList())
            .thenAwait(Duration.ofHours(1))
            .consumeNextWith(l -> System.out.println("flatMap           : " + l))
            .verifyComplete();
    // ... same with concatMap and flatMapSequential
}

@Test
void howLongDoTheyTake() {
    // concatMap: user 1 (3 s + 3 s), then user 2 (2 s + 2 s), then user 3 (1 s + 1 s) = 12 s
    var concat = StepVerifier.withVirtualTime(() -> Flux.range(1, 3).concatMap(OperatorsTest::ordersOf))
            .expectSubscription()
            .thenAwait(Duration.ofSeconds(11)).expectNextCount(5)
            .expectNoEvent(Duration.ofMillis(999))
            .thenAwait(Duration.ofMillis(1)).expectNext("u3-b").verifyComplete();
    // flatMap: all three run at the same time, the slowest (user 1) needs 6 s
    var flat = StepVerifier.withVirtualTime(() -> Flux.range(1, 3).flatMap(OperatorsTest::ordersOf))
            .expectSubscription()
            .thenAwait(Duration.ofSeconds(5)).expectNextCount(5)
            .thenAwait(Duration.ofSeconds(1)).expectNext("u1-b").verifyComplete();
    System.out.println("concatMap needs 12 virtual s, flatMap 6 virtual s; real time: "
            + concat.toMillis() + " ms and " + flat.toMillis() + " ms");
}
```

Output:

```
flatMap           : [u3-a, u2-a, u3-b, u1-a, u2-b, u1-b]
concatMap         : [u1-a, u1-b, u2-a, u2-b, u3-a, u3-b]
flatMapSequential : [u1-a, u1-b, u2-a, u2-b, u3-a, u3-b]
concatMap needs 12 virtual s, flatMap 6 virtual s; real time: 0 ms and 0 ms
```

- `flatMap` subscribes to all inner publishers at once (up to `concurrency`, default 256) and emits **as results arrive**: fastest (time of the slowest, 6 s), order lost.
- `concatMap` finishes each inner publisher before starting the next: order kept, time = sum (12 s).
- `flatMapSequential` runs concurrently (6 s) but re-orders output to the source order.
- Limit fan-out against a fragile service with `flatMap(fn, 8)`: at most 8 calls in flight (e.g. 1,000 ids, 10 at a time: `Flux.fromIterable(ids).flatMap(this::call, 10)`).

Combining

```java
Flux<String> names = Flux.just("Ada", "Linus", "Grace");
Flux<Integer> years = Flux.just(1815, 1969, 1906);
StepVerifier.create(Flux.zip(names, years, (n, y) -> n + "@" + y))
        .expectNext("Ada@1815", "Linus@1969", "Grace@1906").verifyComplete();

StepVerifier.create(Flux.concat(Flux.just(1, 2), Flux.just(3))).expectNext(1, 2, 3).verifyComplete();

Mono<String> fromCache = Mono.empty();
StepVerifier.create(fromCache.switchIfEmpty(Mono.just("from DB")))      // fallback when empty
        .expectNext("from DB").verifyComplete();

StepVerifier.create(Flux.range(1, 7).buffer(3))
        .expectNext(List.of(1, 2, 3), List.of(4, 5, 6), List.of(7)).verifyComplete();
```

`Mono.zip(a, b, c)` is the reactive "call three services in parallel and combine" — `Promise.all` in JavaScript, `CompletableFuture.allOf` in Java.

Common mistakes

- **`map` with an async call**: `flux.map(id -> repo.findById(id))` yields a `Flux<Mono<Book>>` whose inner Monos never run. Use `flatMap`.
- **`flatMap` when order matters** (applying events in sequence) — use `concatMap`.
- **Unbounded `flatMap` against a rate-limited API** — pass a concurrency limit.
- **Business logic in `doOnNext`** — it's for side effects; an exception there becomes an error signal, and it's easy to forget it isn't part of the data flow.

13.5 Testing with `StepVerifier` and virtual time

What it is

`StepVerifier` (in `reactor-test`) subscribes to a publisher and checks its signals step by step: `expectNext(...)`, `expectNextCount(n)`, `assertNext(v -> ...)`, `expectError(Type)`, `expectComplete()`, `thenRequest(n)`, `thenCancel()`. `verify()` (or `verifyComplete()` / `verifyError()`) **runs** the subscription, **blocks** until the script is done, and returns the time taken.

Virtual time replaces Reactor's time-based schedulers with a `VirtualTimeScheduler` whose clock moves only when told (`thenAwait(Duration)`) or jumps when idle. A one-hour interval or a retry with **exponential backoff** (1 s, 2 s, 4 s, ...) is tested in milliseconds, **deterministically**.

How to use virtual time

1. Build the publisher **inside the supplier**: `StepVerifier.withVirtualTime(() -> Flux.interval(...))`, so its time-based operators pick up the virtual scheduler; a publisher created outside keeps real time.
2. Use `expectSubscription()` first, then `expectNoEvent(d)` to assert that nothing happened for a period (it also advances the clock).
3. Use `thenAwait(d)` to move the clock forward, then `expectNext(...)`.

The trap: a test that waits forever

If the script expects a signal that only comes after virtual time passes but never calls `thenAwait`, `verify()` waits **forever** — this chapter's first `howLongDoTheyTake` hung the build until an external alarm killed it. Two defences:

```
# src/test/resources/junit-platform.properties
# safety net: a reactive test that waits for a signal that never comes fails instead of hanging
junit.jupiter.execution.timeout.default=30 s
```

and `verify(Duration.ofSeconds(5))` on real-time scripts (used throughout `webflux-app`).

`StepVerifier.create(flux, 0)` subscribes with initial demand 0, so you drive demand with `thenRequest(n)` (§13.7).

13.6 Error handling

How errors flow

An exception inside an operator becomes an **`onError` signal** that travels down, skipping `map`/`filter`, until an error operator handles it or it reaches the subscriber. **An error terminates the sequence** — later items are never processed (like a rejected Promise, not a `try/catch` inside a loop).

```java
@Test
void anErrorEndsTheStream() {
    StepVerifier.create(Flux.just(1, 2, 0, 4).map(i -> 12 / i))
            .expectNext(12, 6)
            .expectError(ArithmeticException.class)   // 4 is never processed
            .verify();
}

@Test
void fallbacks() {
    Flux<Integer> risky = Flux.just(1, 2, 0, 4).map(i -> 12 / i);
    StepVerifier.create(risky.onErrorReturn(-1)).expectNext(12, 6, -1).verifyComplete();
    StepVerifier.create(risky.onErrorResume(e -> Flux.just(100, 200))).expectNext(12, 6, 100,
200).verifyComplete();
    StepVerifier.create(risky.onErrorMap(e -> new IllegalArgumentException("bad input", e)))
            .expectNext(12, 6).expectErrorMessage("bad input").verify();
}
```

Operator	Effect	Imperative equivalent
`onErrorReturn(v)`	emit a fallback value, then complete	`catch → return v`
`onErrorResume(e -> publisher)`	switch to another publisher (cache, default, another service)	`catch → return other()`
`onErrorMap(e -> new X(e))`	translate the exception	`catch → throw new X(e)`
`doOnError(e -> log)`	side effect, error continues	`catch → log; throw`
`retry(n)`	**re-subscribe** to the source up to n times	loop around the call
`retryWhen(Retry.backoff(...))`	retry with delays, filters, limits	retry library
`timeout(d[, fallback])`	error (or fallback) if no signal within d	`future.get(timeout)`
`onErrorContinue`	drop the failing item and continue (only with supporting operators)	avoid — surprising semantics

Retry with exponential backoff, verified on virtual time

```java
@Test
void retryWithExponentialBackoffOnVirtualTime() {
    List<Long> attemptTimes = new ArrayList<>();
    StepVerifier.withVirtualTime(() -> {
                VirtualTimeScheduler clock = VirtualTimeScheduler.get();
                return Mono.defer(() -> {
                            attemptTimes.add(clock.now(TimeUnit.MILLISECONDS));
                            return Mono.<String>error(new IllegalStateException("down"));
                        })
                        .retryWhen(Retry.backoff(3, Duration.ofSeconds(1)).jitter(0)); // 1 s, 2 s, 4 s
            })
            .thenAwait(Duration.ofMinutes(1))
            .expectErrorMatches(e -> e.getMessage().startsWith("Retries exhausted: 3/3"))
            .verify();
    System.out.println("attempts at virtual ms: " + attemptTimes);
}

@Test
void timeoutWithFallback() {
    StepVerifier.withVirtualTime(() -> Mono.delay(Duration.ofSeconds(5)).map(x -> "slow answer")
                    .timeout(Duration.ofSeconds(2), Mono.just("fallback")))
            .expectSubscription()
            .thenAwait(Duration.ofSeconds(2))
            .expectNext("fallback")
            .verifyComplete();
    ...
}
```

Output:

```
attempts at virtual ms: [0, 1000, 3000, 7000]
retry(5): succeeded after 3 attempts
timeout(2s) on a 5 s call -> fallback / TimeoutException
onErrorReturn / onErrorResume / onErrorMap verified
```

Attempt 1 runs at 0; retries wait 1 s, 2 s, 4 s (1,000, 3,000, 7,000 ms); then `Retries exhausted: 3/3`. Real systems keep the default **jitter** (random ±50 % here) so clients don't retry in lockstep; `jitter(0)` is for the deterministic test. Retry and circuit-breaker (stop calling a failing service for a while) design: microservices notes — retry.

Retry re-subscribes, which repeats work only for a lazy source (`Mono.defer(...)`, `fromCallable`, `WebClient` calls); `Mono.just(valueComputedEarlier)` just replays the value.

Common mistakes

- **Retrying non-idempotent calls** (*idempotent*: same effect once or many times; a card-charging `POST` is not) without an idempotency key (a unique id the server uses to ignore duplicates; System Design Fundamentals).
- **Retrying on 4xx** (client errors fail again). Filter to transient errors (e.g. 503): `Retry.backoff(...).filter(e -> e instanceof ServiceUnavailable)`.
- `try/catch` **around a reactive chain** catches only assembly-time errors, not those raised when the pipeline runs.

13.7 Backpressure in practice

What it is

Backpressure = the consumer tells the producer how much it can handle via `request(n)`. Most Reactor operators pass demand **upstream** (towards the source; *downstream* is towards the subscriber) automatically; `block()` or WebFlux's HTTP writer request in batches.

Demand you can see — `BackpressureTest`

```
@Test
void subscriberControlsTheRate() {
    List<Long> requests = new CopyOnWriteArrayList<>();
    List<Integer> received = new ArrayList<>();

    Flux.range(1, 10)
        .doOnRequest(requests::add)                 // what the subscriber asked for
        .subscribe(new BaseSubscriber<Integer>() {
            @Override protected void hookOnSubscribe(Subscription s) { request(2); }  // start with 2
            @Override protected void hookOnNext(Integer v) {
                received.add(v);
                if (received.size() % 2 == 0) request(2);                           // then 2 at a time
            }
        });
    System.out.println("requests = " + requests + "\nreceived = " + received);
}

@Test
void limitRateReplenishesInBatches() {
    List<Long> upstreamRequests = new CopyOnWriteArrayList<>();
    Flux.range(1, 100)
        .doOnRequest(upstreamRequests::add)
        .limitRate(10)                              // never ask upstream for more than 10
        .blockLast();                               // downstream asks for "unbounded"
    System.out.println("limitRate(10) upstream requests = " + upstreamRequests);
}

@Test
void stepVerifierCanRequestManually() {
    StepVerifier.create(Flux.range(1, 5), 0)          // initial demand 0
            .expectSubscription()
            .expectNoEvent(Duration.ofMillis(50))     // nothing arrives without demand
            .thenRequest(2).expectNext(1, 2)
            .thenRequest(3).expectNext(3, 4, 5)
            .verifyComplete();
}
```

Output:

```
limitRate(10) upstream requests = [10, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8]
demand 0 -> nothing; request(2) -> 1,2; request(3) -> 3,4,5
requests = [2, 2, 2, 2, 2, 2]
received = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

- The source sent exactly as many items as requested, 2 at a time; with demand 0 a well-behaved publisher sends nothing.
- `limitRate(10)` turns a downstream "everything" (`Long.MAX_VALUE`) into bounded requests: 10 first, then **8 more when 75 % are consumed** (default "low tide"), so the next batch is in flight before the buffer runs dry. Use it to protect paged sources (a database cursor walking results in batches, a paged API).

When the source can't slow down: overflow strategies

Some sources **ignore demand** (UI events, sensors, a price ticker, a hot message stream); you choose what happens to the excess. The test's `ticker()` pushes 1..20 at once through `Flux.create(..., OverflowStrategy.IGNORE)` while the consumer asks for 5:

```java
static Flux<Integer> ticker() {
    return Flux.create(sink -> {
        for (int i = 1; i <= 20; i++) sink.next(i);
        sink.complete();
    }, FluxSink.OverflowStrategy.IGNORE);   // IGNORE = no built-in protection, the operator below decides
}

StepVerifier.create(ticker().onBackpressureDrop(dropped::add), 5).expectNext(1, 2, 3, 4, 5).verifyComplete();

SlowConsumer latest = new SlowConsumer();              // requests 5 on subscribe
ticker().onBackpressureLatest().subscribe(latest);     // 20 items pushed during this call
latest.request(5);                                     // ask for more afterwards

SlowConsumer buffered = new SlowConsumer();
ticker().onBackpressureBuffer(10).subscribe(buffered); // 6..15 fit in the buffer, 16 overflows
buffered.request(50);
```

Output:

```
DROP   : kept 1..5, dropped 15 items (6..20)
LATEST : [1, 2, 3, 4, 5, 20] complete
BUFFER(10): [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15] error: The receiver is overrun by more signals
than expected (bounded queue...)
```

Strategy	What happens to excess	Use for
`onBackpressureLatest`	keep only the newest waiting item	"current price" (a feed faster than the UI renders), UI state
`onBackpressureBuffer(n)`	queue up to n, then **error** (or a chosen overflow action)	short bursts that must not be lost
`onBackpressureBuffer()` (unbounded)	queue everything	risky: a consumer slower than the producer grows it until `OutOfMemoryError`
`onBackpressureError`	fail immediately	when overflow is a bug
`onBackpressureError`	fail immediately	when overflow is a bug

`Flux.create` accepts the same choices directly (`OverflowStrategy.DROP/LATEST/BUFFER/ERROR`).

Backpressure across the network

Inside one JVM demand is exact; across HTTP it becomes **TCP flow control** (TCP stops the sender when the receiver's buffer is full; System Design Fundamentals). When WebFlux writes a `Flux` to a slow client, the socket send buffer fills, Netty stops requesting, and demand stops at the source (e.g. the R2DBC cursor). Between services HTTP/1.1 has no message-level backpressure; RSocket or gRPC (binary protocols with flow control), or a broker in between, provide it.

13.8 Schedulers and threads

What it is

Reactor is **concurrency-agnostic**: each signal runs on whatever thread emitted it (subscriber, Netty event loop, timer). **Schedulers** move work onto specific pools:

Scheduler	Threads	For
`Schedulers.parallel()`	one per CPU core	CPU-bound work, timers (`interval`, `delayElements`)
`Schedulers.boundedElastic()`	grows on demand up to 10 × cores (default), idle threads reclaimed	**wrapping blocking calls** (JDBC, legacy SDKs, file I/O)
`Schedulers.single()`	one reusable thread	low-latency one-off tasks
`Schedulers.immediate()`	the current thread	tests, defaults
Reactor Netty event loops (`reactor-http-nio-N`)	one per core (min 4)	network I/O — **never block these**

- **`publishOn(s)`** switches the thread for everything **after** it in the chain.
- **`subscribeOn(s)`** chooses the thread on which the **subscription happens**, moving the **source** (and everything up to the first `publishOn`); its position in the chain doesn't matter.

```
@Test
void publishOnSwitchesDownstreamSubscribeOnMovesTheSource() {
    List<String> log = new CopyOnWriteArrayList<>();
    Flux.just(1)
        .doOnNext(i -> log.add("source    on " + thread()))
        .publishOn(Schedulers.parallel())               // everything BELOW runs on parallel
        .doOnNext(i -> log.add("after pub on " + thread()))
        .subscribeOn(Schedulers.boundedElastic())       // the subscription (and source) runs here
        .blockLast();
    log.forEach(System.out::println);
}

@Test
void wrapABlockingCall() {
    // A blocking API (JDBC, legacy SDK) must not run on the few event-loop/parallel threads.
    Mono<String> legacy = Mono.fromCallable(() -> {
                Thread.sleep(50);                       // pretend: blocking I/O
                return "legacy result on " + thread();
            })
            .subscribeOn(Schedulers.boundedElastic());   // dedicated, growable pool for blocking work
    System.out.println(legacy.block());
}
```

Output (thread numbers replaced by `N`):

```
legacy result on boundedElastic-N
source    on boundedElastic-N
after pub on parallel-N
```

`Mono.fromCallable(blocking).subscribeOn(Schedulers.boundedElastic())` is how to call unavoidable blocking code from a reactive app: event loops stay free, at the cost of a thread per concurrent call — exactly what reactive meant to avoid. If most calls block, a reactive stack buys nothing. **`BlockHound`** (a Java agent from the Reactor team) detects blocking calls on non-blocking threads in tests.

13.9 Hot vs cold, sinks and context

Cold vs hot

- **Cold** (default): each subscriber gets its own run from the start (a query, a `WebClient` call, `Flux.range`).
- **Hot**: data is produced whether or not anyone listens; late subscribers **miss** earlier items (stock ticks, mouse events, a chat room).

Sinks (`Sinks.many()`, `Sinks.one()`) are the safe way to push values into a `Flux` from **imperative** code, e.g. broadcasting events to all SSE clients:

```
Sinks.Many<String> prices = Sinks.many().multicast().directBestEffort();
prices.asFlux().subscribe(early::add);
prices.tryEmitNext("p1");
prices.asFlux().subscribe(late::add);   // joins after p1 was emitted
prices.tryEmitNext("p2");
prices.tryEmitComplete();
```

Output:

```
early subscriber: [p1, p2], late subscriber: [p2]
```

Other sinks: `Sinks.many().replay().limit(n)` (late subscribers get the last n), `unicast()` (one subscriber, buffers). Make a cold publisher hot with `share()`, `publish().autoConnect()`, or `cache()` (replays to later subscribers).

Context: the reactive replacement for `ThreadLocal`

A pipeline may hop threads, so `ThreadLocal` values (MDC request id, security context, current transaction) get lost. Reactor's `Context` is an immutable key/value map attached to the **subscription**: written at the bottom with `contextWrite`, read anywhere above with `deferContextual`:

```
Mono<String> greeting = Mono.deferContextual(ctx -> Mono.just("request " + ctx.get("requestId")));
StepVerifier.create(greeting.contextWrite(c -> c.put("requestId", "r-42")))
        .expectNext("request r-42").verifyComplete();
```

Output:

```
context value read inside the pipeline: r-42
```

Spring uses it for you: reactive Spring Security stores the `SecurityContext` there, reactive transactions keep the connection there. For MDC and **tracing** (following a request across threads and services, Chapter 15), Boot bridges `ThreadLocal`s and the Reactor `Context` via Micrometer's context-propagation library (`spring.reactor.context-propagation=auto`).

13.10 Spring WebFlux: annotated controllers

What it is

Spring WebFlux (Framework 5.0+) is the reactive counterpart of Spring MVC, on **Reactor Netty** by default (Tomcat and Jetty work via the Servlet non-blocking I/O API). The flow mirrors MVC: `HttpHandler` → `WebFilter`s → **`DispatcherHandler`** → `HandlerMapping` → `HandlerAdapter` → your handler → `HttpMessageWriter` (Jackson 3 **codecs**: byte ↔ object encoders/decoders), all non-blocking.

	Spring MVC	**Spring WebFlux**
I/O model	blocking (one thread per request) or virtual threads	non-blocking event loop
Server	Tomcat / Jetty (Servlet)	Reactor Netty (default), Tomcat, Jetty
Return types	`T`, `ResponseEntity<T>` (also accepts `Mono`/`Flux`)	`Mono<T>`, `Flux<T>`, `T`
Data access	JDBC, JPA	R2DBC, reactive MongoDB/Redis/Cassandra
HTTP client	`RestClient`	`WebClient`
Test client	`MockMvcTester`, `RestTestClient` (ch12)	`WebTestClient`
Starter (Boot 4)	`spring-boot-starter-webmvc`	`spring-boot-starter-webflux`

If **both** starters are on the classpath, Boot starts **MVC**. Force WebFlux with `spring.main.web-application-type=reactive`.

`webflux-app/pom.xml` (main parts):

```
<!-- Spring WebFlux + Reactor Netty server + Jackson 3 -->
<dependency><groupId>org.springframework.boot</groupId><artifactId>spring-boot-starter-webflux</artifactId>
</dependency>
<!-- Boot 4: the auto-configured WebClient.Builder lives in its own module/starter -->
<dependency><groupId>org.springframework.boot</groupId><artifactId>spring-boot-starter-webclient</artifactId>
</dependency>
<!-- Spring Data R2DBC + the reactive H2 driver -->
<dependency><groupId>org.springframework.boot</groupId><artifactId>spring-boot-starter-data-r2dbc</artifactId>
</dependency>
<dependency><groupId>io.r2dbc</groupId><artifactId>r2dbc-h2</artifactId><scope>runtime</scope></dependency>
<!-- tests: @WebFluxTest, WebTestClient, reactor-test / @DataR2dbcTest -->
<dependency><groupId>org.springframework.boot</groupId><artifactId>spring-boot-starter-webflux-test</artifactId>
<scope>test</scope></dependency>
<dependency><groupId>org.springframework.boot</groupId><artifactId>spring-boot-starter-data-r2dbc-
test</artifactId><scope>test</scope></dependency>
```

> **Boot 4 gotcha (found while building this chapter):** `spring-boot-starter-webflux` no longer brings the `WebClient.Builder` auto-configuration. Without `spring-boot-starter-webclient` the app fails with *"No qualifying bean of type 'org.springframework.web.reactive.function.client.WebClient$Builder' available"*.

The controller

Same annotations as MVC (Chapter 5); only the return types change:

```
@RestController
@RequestMapping("/api/books")
public class BookController {
    private final BookRepository repo;                // reactive repository (§13.14)

    public BookController(BookRepository repo) { this.repo = repo; }

    @GetMapping
    public Flux<Book> all(@RequestParam(required = false) String q) {
        return q == null ? repo.findAll() : repo.search(q);
    }

    @GetMapping("/{id}")
    public Mono<Book> one(@PathVariable long id) {
        return repo.findById(id)
                .switchIfEmpty(Mono.error(new ResponseStatusException(HttpStatus.NOT_FOUND, "no book " + id)));
    }

    @PostMapping
    public Mono<ResponseEntity<Book>> create(@RequestBody Mono<Book> body) {
        return body.map(b -> b.withId(null))                    // never trust a client-sent id
                .flatMap(repo::save)
                .map(saved -> ResponseEntity.created(URI.create("/api/books/" + saved.id())).body(saved));
    }
    ...
}
```

- `Flux<Book>` is written as a JSON array (or streamed as NDJSON — one JSON object per line — / SSE when the client asks for `application/x-ndjson` / `text/event-stream`).
- An empty `Mono` would give `200` with an empty body; `switchIfEmpty(Mono.error(...))` makes "not found" a 404.
- `@RequestBody Mono<Book>` makes even body parsing non-blocking (plain `@RequestBody Book` also works).
- **You return the pipeline, the framework subscribes** — no `subscribe()` or `block()` anywhere.

Testing it — `@WebFluxTest` + `WebTestClient`

```java
/** WebFlux slice: controller + codecs + error handling; the repository is a mock returning Mono/Flux. */
@WebFluxTest(BookController.class)
class BookControllerWebFluxTest {

    @Autowired WebTestClient client;
    @MockitoBean BookRepository repo;

    static final Book DDD = new Book(1L, "Domain-Driven Design", "Eric Evans", "9780321125217");

    @Test
    void listReturnsJsonArray() {
        given(repo.findAll()).willReturn(Flux.just(DDD));
        client.get().uri("/api/books").exchange()
              .expectStatus().isOk()
              .expectHeader().contentType(MediaType.APPLICATION_JSON)
              .expectBodyList(Book.class).hasSize(1).contains(DDD);
    }

    @Test
    void missingBookIs404() {
        given(repo.findById(99L)).willReturn(Mono.empty());
        client.get().uri("/api/books/99").exchange()
              .expectStatus().isNotFound()
              .expectBody()
              .consumeWith(r -> System.out.println("GET /api/books/99 -> 404 " + new String(r.getResponseBody())))
              .jsonPath("$.status").isEqualTo(404);
    }

    @Test
    void createReturns201() {
        given(repo.save(new Book(null, "Refactoring", "Martin Fowler", "9780134757599")))
                .willReturn(Mono.just(new Book(7L, "Refactoring", "Martin Fowler", "9780134757599")));
        client.post().uri("/api/books").contentType(MediaType.APPLICATION_JSON)
              .bodyValue("""
                      {"id":123,"title":"Refactoring","author":"Martin Fowler","isbn":"9780134757599"}""")
              .exchange()
              .expectStatus().isCreated()
              .expectHeader().location("/api/books/7")
              .expectBody().jsonPath("$.id").isEqualTo(7);
    }
}
```

Output:

```
POST /api/books (client id 123 ignored) -> 201 Location: /api/books/7
GET /api/books/99 -> 404 {"timestamp":"2026-09-29T00:24:18.499Z","path":"/api/books/99","status":404,"error":"Not Found","requestId":"25c24062"}
GET /api/books -> 200 [Domain-Driven Design]
```

- `@WebFluxTest` is the WebFlux slice (ch12 §12.7); unlike `@SpringBootTest` it provides a `WebTestClient` without an extra annotation.
- Mocked reactive repositories must return `Mono`/`Flux`, never `null` (→ `NullPointerException` inside the framework).
- The 404 body is Boot's default reactive error format; for RFC 9457 Problem Details set `spring.webflux.problemdetails.enabled=true` or add a `@RestControllerAdvice` returning `ProblemDetail`, as in MVC.

13.11 Functional endpoints

What it is

A second model (also in MVC via `RouterFunctions` in `spring-webmvc`): routes are **code** — a `RouterFunction<ServerResponse>` bean — and handlers are plain functions `ServerRequest → Mono<ServerResponse>`. No annotations or reflection: explicit, easily unit-tested routing.

```
@Configuration(proxyBeanMethods = false)
public class BookRoutes {

    @Bean
    RouterFunction<ServerResponse> bookRouter(BookHandler h) {
        return route()
                .path("/fn/books", b -> b
                        .GET("/{isbn}", h::byIsbn)
                        .GET("", h::byAuthor))                      // /fn/books?author=...
                .filter((req, next) -> {                           // a filter for this router only
                    req.exchange().getResponse().getHeaders().add("X-Router", "functional");
                    return next.handle(req);
                })
                .build();
    }

    @Component
    static class BookHandler {
        private final BookRepository repo;
        BookHandler(BookRepository repo) { this.repo = repo; }

        Mono<ServerResponse> byIsbn(ServerRequest req) {
            return repo.findByIsbn(req.pathVariable("isbn"))
                    .flatMap(b -> ServerResponse.ok().contentType(MediaType.APPLICATION_JSON).bodyValue(b))
                    .switchIfEmpty(ServerResponse.notFound().build());
        }

        Mono<ServerResponse> byAuthor(ServerRequest req) {
            return req.queryParam("author")
                    .map(a -> ServerResponse.ok().body(repo.findByAuthor(a), Book.class))
                    .orElseGet(() -> ServerResponse.badRequest().bodyValue("author is required"));
        }
    }
}
```

Found while testing: the first filter changed the returned `ServerResponse`'s headers (`res.headers().add(...)`) and failed with `UnsupportedOperationException` (500) — a built `ServerResponse` is immutable. Add headers through the exchange (as above) or build a new response.

Testing without Spring

Routes are plain objects, so `WebTestClient` binds straight to them — no context, no server:

```
class FunctionalRoutesTest {
    BookRepository repo = mock(BookRepository.class);
    WebTestClient client = WebTestClient
            .bindToRouterFunction(new BookRoutes().bookRouter(new BookRoutes.BookHandler(repo)))
            .build();

    @Test
    void byIsbnFoundAndNotFound() {
        given(repo.findByIsbn("111")).willReturn(Mono.just(new Book(1L, "T", "A", "111")));
        given(repo.findByIsbn("222")).willReturn(Mono.empty());

        client.get().uri("/fn/books/111").exchange()
              .expectStatus().isOk()
              .expectHeader().valueEquals("X-Router", "functional")
              .expectBody().jsonPath("$.title").isEqualTo("T");
        client.get().uri("/fn/books/222").exchange().expectStatus().isNotFound();
    }
    ...
}
```

Output:

```
GET /fn/books/111 -> 200 (X-Router: functional), /fn/books/222 -> 404
GET /fn/books?author=Evans -> 200 [1 book], /fn/books -> 400 author is required
```

Annotated vs functional

	Annotated controllers	Functional endpoints
Routing	annotations, discovered by scanning	explicit code, composable
Argument binding / validation	automatic (`@PathVariable`, `@Valid`)	manual (`req.pathVariable`, `req.bodyToMono`, call a `Validator`)
Familiar to most Spring devs	yes	less
Good for	most CRUD APIs	small services, dynamic routes, tight control, GraalVM-friendly code (ahead-of-time native executables)

Both can live in one app (as here).

13.12 Full-stack WebFlux tests and Server-Sent Events

Server-Sent Events

Server-Sent Events (SSE) is a simple standard for a server to **push** text events over one long-lived HTTP response (`Content-Type: text/event-stream`); browsers consume it with `EventSource`, reconnecting automatically. One-way (server → client); for two-way use WebSockets (an HTTP connection upgraded to a long-lived two-way channel). In WebFlux an SSE endpoint is just a `Flux` with that media type:

```
/** Server-Sent Events: one "tick" per interval, pushed over a single open HTTP response. */
@GetMapping(path = "/ticks", produces = MediaType.TEXT_EVENT_STREAM_VALUE)
public Flux<String> ticks(@RequestParam(defaultValue = "1000") long everyMs) {
    return Flux.interval(Duration.ofMillis(everyMs)).map(i -> "tick " + i);
}
```

Each element is written as `data:tick 0` plus a blank line; for event names, ids or retry hints return `Flux<ServerSentEvent<T>>`. When the client disconnects the subscription is **cancelled** and `Flux.interval` stops — no leaked timers. (Spring MVC offers `SseEmitter` or returning a `Flux`, but each open stream then holds a servlet async request.)

The full-stack test

```
/** Real Netty server on a random port, real R2DBC/H2 database seeded by data.sql. */
@SpringBootTest(webEnvironment = SpringBootTest.WebEnvironment.RANDOM_PORT)
@AutoConfigureWebTestClient
class FullStackWebFluxTest {

    @Autowired WebTestClient client;

    @Test
    void searchHitsTheDatabase() {
        client.get().uri("/api/books?q=reactive").exchange()
              .expectStatus().isOk()
              .expectBody()
              .consumeWith(r -> System.out.println("GET /api/books?q=reactive -> " + new
String(r.getResponseBody())))
              .jsonPath("$.length()").isEqualTo(2);
    }

    @Test
    void serverSentEvents() {
        Flux<String> events = client.get().uri("/api/books/ticks?everyMs=100")
                .accept(MediaType.TEXT_EVENT_STREAM)
                .exchange()
                .expectStatus().isOk()
                .expectHeader().contentTypeCompatibleWith(MediaType.TEXT_EVENT_STREAM)
                .returnResult(String.class)
                .getResponseBody();                      // a live Flux of the streamed events

        StepVerifier.create(events)
                .expectNext("tick 0", "tick 1", "tick 2")
                .thenCancel()                            // closes the connection; the server stops its interval
                .verify(Duration.ofSeconds(5));
    }
}
```

Output:

```
GET /api/books?q=reactive -> [{"id":2,"title":"Hands-On Reactive Programming in Spring 5","author":"Oleh
Dokuka","isbn":"9781787284951"},{"id":1,"title":"Reactive Spring","author":"Josh Long","isbn":"9781732910409"}]
SSE received: tick 0, tick 1, tick 2 (then cancelled)
```

- In Boot 4, `WebTestClient` in a `@SpringBootTest` needs **`@AutoConfigureWebTestClient`** (from `spring-boot-starter-webflux-test`); it is then bound to the random-port server.
- `returnResult(String.class).getResponseBody()` gives the **live stream**; `WebTestClient` + `StepVerifier` (take a few items, cancel) is the standard way to test infinite/streaming endpoints.

13.13 `WebClient`: calling other services without blocking

What it is

`WebClient` is the non-blocking reactive HTTP client (Framework 5.0+). Boot auto-configures a `WebClient.Builder` with codecs and observability (per-call metrics and tracing, Chapter 15) — from `spring-boot-starter-webclient` in Boot 4. Its API mirrors `RestClient` (Chapter 6) but returns `Mono`/`Flux`.

```java
@Component
public class InventoryClient {
    public record Stock(String isbn, int copies) { }

    private final WebClient http;

    public InventoryClient(WebClient.Builder builder, @Value("${inventory.base-url}") String baseUrl) {
        this.http = builder.baseUrl(baseUrl).build();
    }

    public Mono<Stock> stock(String isbn) {
        return http.get().uri("/stock/{isbn}", isbn)
                .retrieve()
                .onStatus(HttpStatusCode::is5xxServerError,
                        r -> Mono.error(new InventoryUnavailable("inventory said " + r.statusCode().value())))
                .bodyToMono(Stock.class)
                .timeout(Duration.ofMillis(500))                                    // per attempt
                .retryWhen(Retry.backoff(2, Duration.ofMillis(100))
                        .filter(e -> e instanceof InventoryUnavailable));          // only retry 5xx
    }

    /** Several calls at once: total time is roughly the slowest call, not the sum. */
    public Mono<Integer> totalCopies(List<String> isbns) {
        return Flux.fromIterable(isbns)
                .flatMap(this::stock)
                .map(Stock::copies)
                .reduce(0, Integer::sum);
    }
}
```

- `retrieve()` + `bodyToMono` is the simple path; 4xx/5xx become `WebClientResponseException` unless mapped with `onStatus`.
- `timeout` **before** `retryWhen` limits **each attempt**; after it, the whole operation.
- Only `InventoryUnavailable` (5xx) is retried, not a 404.

Tested against a real local server

The test starts a real **Reactor Netty HTTP server** on a random port, scripted to be fast, slow, flaky or down; `@DynamicPropertySource` points the client at it:

```java
@SpringBootTest(webEnvironment = SpringBootTest.WebEnvironment.NONE)
class InventoryClientTest {

    static final AtomicInteger flakyCalls = new AtomicInteger();
    static final AtomicInteger downCalls = new AtomicInteger();

    static final DisposableServer INVENTORY = HttpServer.create()
            .host("127.0.0.1").port(0)
            .route(r -> r.get("/stock/{isbn}", (req, res) -> {
                String isbn = req.param("isbn");
                String json = "{\"isbn\":\"" + isbn + "\",\"copies\":3}";
                return switch (isbn) {
                    case "slow" -> res.header("Content-Type", "application/json")
                            .sendString(Mono.just(json).delayElement(Duration.ofSeconds(2)));
                    case "flaky" -> flakyCalls.incrementAndGet() <= 2
                            ? res.status(503).send()
                            : res.header("Content-Type", "application/json").sendString(Mono.just(json));
                    case "down" -> { downCalls.incrementAndGet(); yield res.status(500).send(); }
                    default -> res.header("Content-Type", "application/json")      // 300 ms "network + work"
                            .sendString(Mono.just(json).delayElement(Duration.ofMillis(300)));
                };
            }))
            .bindNow();

    @DynamicPropertySource
    static void props(DynamicPropertyRegistry r) {
        r.add("inventory.base-url", () -> "http://127.0.0.1:" + INVENTORY.port());
    }

    @Autowired InventoryClient client;

    @Test
    void callsRunConcurrently() {
        long t0 = System.nanoTime();
        Integer total = client.totalCopies(List.of("a", "b", "c", "d", "e")).block();
        long ms = (System.nanoTime() - t0) / 1_000_000;
        System.out.println("5 calls of 300 ms each -> total copies " + total + " in " + ms + " ms (machine-
dependent)");
    }

    @Test
    void timeoutFires() {

StepVerifier.create(client.stock("slow")).expectError(TimeoutException.class).verify(Duration.ofSeconds(5));
    }

    @Test
    void retriesTransient5xx() {
        StepVerifier.create(client.stock("flaky")).expectNextCount(1).verifyComplete();
        StepVerifier.create(client.stock("down"))
                .expectErrorMatches(e -> e.getMessage().startsWith("Retries exhausted: 2/2"))
                .verify(Duration.ofSeconds(5));
        assertThat(downCalls).hasValue(3);   // 1 try + 2 retries
    }
}
```

Output:

```
5 calls of 300 ms each -> total copies 15 in 322 ms (machine-dependent)
slow (2 s) -> TimeoutException after the 500 ms timeout
flaky: 503, 503, 200 -> success after 3 calls; down: gave up after 3 calls
```

Five 300 ms calls finished in **322 ms**, not 1,500 ms: `flatMap` ran them concurrently and no thread waited. (`RestClient` on virtual threads gets the same wall time — compared in §13.15.)

Common mistakes

- **`.block()` on a `WebClient` call inside a WebFlux handler** → `IllegalStateException`. Return the `Mono`.

- **`WebClient.create()` per call** → no shared **connection pool** (reused open connections), no Boot customisation. Build once from the injected builder.
- **Ignoring an error response's body** → the connection may not be released. Let `retrieve()` handle it, or consume/release the body in `exchangeToMono`.
- **No timeout** → a hung remote service keeps the request open forever. Add `timeout(...)` and/or connect/response timeouts on the underlying `HttpClient`.

13.14 R2DBC: reactive SQL

What it is

JDBC is **blocking by design**: `executeQuery()` holds the thread until rows arrive. **R2DBC** (Reactive Relational Database Connectivity, spec 1.0 in 2022) is a non-blocking SPI (service provider interface: a standard API each vendor's driver implements) for SQL databases, with drivers for PostgreSQL, MySQL/MariaDB, SQL Server, Oracle and H2. **Spring Data R2DBC** adds repositories and `R2dbcEntityTemplate`; `DatabaseClient` offers fluent SQL.

It is **not an ORM** (like Hibernate): no lazy loading, relationships, cascades, dirty checking (auto-detecting changed fields) or first-level cache. One entity = one table; joins are hand-written. Deliberately — lazy loading would mean hidden blocking I/O. Compare Chapter 8.

Configuration

```
spring.r2dbc.url=r2dbc:h2:mem:///books;DB_CLOSE_DELAY=-1
# run schema.sql / data.sql through R2DBC at startup
spring.sql.init.mode=always
```

For PostgreSQL: `spring.r2dbc.url=r2dbc:postgresql://host:5432/db` with `org.postgresql:r2dbc-postgresql`. Boot adds a pool (`r2dbc-pool`) automatically. Flyway and Liquibase migrations need JDBC, so apps usually keep a JDBC driver *only* for migrations.

Entity and repository

```
/**
 * A row of table BOOK. id == null means "not saved yet" (INSERT), otherwise UPDATE.
 * No explicit name: Spring Data R2DBC quotes identifiers, so @Table("book") would look for a
 * lower-case "book" table, while H2 stored the unquoted name as BOOK.
 */
@Table
public record Book(@Id Long id, String title, String author, String isbn) {
    public Book withId(Long newId) { return new Book(newId, title, author, isbn); }
}

public interface BookRepository extends ReactiveCrudRepository<Book, Long> {

    Flux<Book> findByAuthor(String author);            // derived query, like Spring Data JPA

    Mono<Book> findByIsbn(String isbn);

    @Query("SELECT * FROM book WHERE LOWER(title) LIKE LOWER(CONCAT('%', :word, '%')) ORDER BY title")
    Flux<Book> search(String word);
}
```

The Javadoc records a real failure: with `@Table("book")` every statement failed with *`Table "book" not found (candidates are: "BOOK")`*. Spring Data R2DBC **quotes** identifiers (case-sensitive), and the unquoted `CREATE TABLE book` produced `BOOK` in H2 (PostgreSQL folds to lower case instead). Omit the name, so Spring derives it in the dialect's default case, or match the case exactly.

Tests — `@DataR2dbcTest`, `DatabaseClient`, a reactive transaction

```
/** R2DBC slice: repositories + DatabaseClient on H2. Reactive tests are NOT rolled back automatically. */
@DataR2dbcTest
class BookRepositoryR2dbcTest {

    @Autowired BookRepository repo;
    @Autowired DatabaseClient db;
    @Autowired ReactiveTransactionManager txManager;

    @Test
    void derivedAndCustomQueries() {
        StepVerifier.create(repo.findByAuthor("Brian Goetz").map(Book::title))
                .expectNext("Java Concurrency in Practice").verifyComplete();
        StepVerifier.create(repo.search("SPRING").map(Book::title).collectList())
                .assertNext(l -> System.out.println("search('SPRING') -> " + l))
                .verifyComplete();
    }

    @Test
    void saveAssignsId() {
        StepVerifier.create(repo.save(new Book(null, "Release It!", "Michael Nygard", "9781680502398"))
                        .flatMap(saved -> repo.findById(saved.id())))
                .assertNext(b -> System.out.println("saved and re-read: " + b))
                .verifyComplete();
        repo.deleteAll(repo.findByIsbn("9781680502398")).block();   // clean up ourselves
    }

    @Test
    void databaseClientAndReactiveTransaction() {
        TransactionalOperator tx = TransactionalOperator.create(txManager);

        Mono<Long> insertThenFail = db.sql("INSERT INTO book(title, author, isbn) VALUES ('Tmp', 'X', 'tmp-1')")
                .fetch().rowsUpdated()
                .then(Mono.<Long>error(new IllegalStateException("fail after insert")))
                .as(tx::transactional);                      // the whole pipeline is one transaction

        StepVerifier.create(insertThenFail).expectErrorMessage("fail after insert").verify();

        StepVerifier.create(db.sql("SELECT COUNT(*) AS n FROM book WHERE isbn = 'tmp-1'")
                        .map(row -> row.get("n", Long.class)).one())
                .assertNext(n -> System.out.println("rows with isbn tmp-1 after rollback: " + n))
                .verifyComplete();
    }
}
```

Output:

```
saved and re-read: Book[id=7, title=Release It!, author=Michael Nygard, isbn=9781680502398]
rows with isbn tmp-1 after rollback: 0
search('SPRING') -> [Hands-On Reactive Programming in Spring 5, Reactive Spring]
```

- **Derived queries and `@Query`** work as in Spring Data JPA, returning `Flux`/`Mono`. `save` with `id == null` INSERTs and returns the row with its generated id (7, because the in-memory DB is shared with other test contexts and `data.sql` ran several times).
- **Reactive transactions**: `TransactionalOperator` (or `@Transactional` on a method returning `Mono`/`Flux`, backed by `R2dbcTransactionManager`) binds the connection to the **subscription's Context**, not a thread (Chapter 7 has the thread-bound version). The error after the INSERT rolled it back: 0 rows.
- **`@DataR2dbcTest` is not transactional**: unlike `@DataJpaTest` it isn't meta-annotated `@Transactional` (checked in Boot 4.1.1), because test-managed transactions are thread-bound. Clean up in the test (`saveAssignsId`).

When not to use R2DBC

When you need JPA features (relationships, lazy loading, caching), the team knows JPA well, or you're on virtual threads anyway. The ecosystem is smaller and some drivers lag their JDBC versions.

13.15 Virtual threads vs WebFlux: a measured comparison

The question

Java 21 made **virtual threads** final (Java ch15): JVM-managed threads costing a few hundred bytes to a few KB instead of a platform thread's MB-sized stack. A virtual thread blocking on I/O is **unmounted** from its carrier (platform) thread, which runs another one — event-loop-like scalability *without code changes*. Spring Boot (3.2+) switches Tomcat request handling, `@Async`, `@Scheduled` and more to virtual threads with **one property**:

```
spring.threads.virtual.enabled=true
```

For an I/O-bound endpoint, how do **MVC on platform threads**, **MVC on virtual threads** and **WebFlux** compare?

The setup

Two tiny projects share one endpoint: wait 100 ms (simulated I/O), then answer.

```
// load-mvc: blocking. The thread WAITS 100 ms, like a JDBC query or a blocking HTTP call.
@GetMapping("/io")
String io(@RequestParam(defaultValue = "100") long ms) throws InterruptedException {
    Thread.sleep(ms);
    return "ok on " + (Thread.currentThread().isVirtual() ? "virtual" : "platform") + " thread";
}

// load-webflux: non-blocking. A timer completes the response after 100 ms; no thread waits.
@GetMapping("/io")
Mono<String> io(@RequestParam(defaultValue = "100") long ms) {
    return Mono.delay(Duration.ofMillis(ms)).map(x -> "ok (non-blocking)");
}

// load-webflux: ANTI-PATTERN. Blocking inside a reactive handler parks an event-loop thread.
@GetMapping("/io-blocking")
Mono<String> ioBlocking(@RequestParam(defaultValue = "100") long ms) throws InterruptedException {
    Thread.sleep(ms);
    return Mono.just("ok (blocked " + ... + ")");
}
```

- `load-mvc` runs twice: `spring.threads.virtual.enabled=false` (Tomcat's 200 platform workers) and `=true`, each a separate `@SpringBootTest(webEnvironment = RANDOM_PORT)` class marked `@DirtiesContext` so its server stops before the next measurement.
- The load generator (`LoadGenerator.java`) is a **closed loop**: `concurrency` clients, each a virtual thread with the JDK `HttpClient` (HTTP/1.1), send requests back-to-back. It reports wall time, throughput (req/s), p50 (median) and p99 (99% faster) latency, and the JVM's **peak platform thread count** (`ThreadMXBean`, excluding virtual threads).
- An unreported warm-up runs first (JIT compilation, connection setup).
- **Caveats:** client and server share one JVM and laptop; `Thread.sleep`/`Mono.delay` stand in for real I/O; no database pool or downstream to saturate. The numbers show the *shape* of the difference, not production behaviour.

Results (machine-dependent)

Real output (14-core Apple M4 Pro, JDK 26 running Java 25 bytecode):

```
MVC, spring.threads.virtual.enabled=false -> "ok on platform thread"
| mode                         | conc. |   reqs | wall ms |  req/s | p50 ms | p99 ms | peak platform threads |
errors |
| MVC Platform threads         |   100 |   2000 |    2157 |    927 |    105 |    119 |                   231 |
0 |
| MVC Platform threads         |  1000 |  10000 |    5238 |   1909 |    517 |    535 |                   231 |
0 |
MVC, spring.threads.virtual.enabled=true -> "ok on virtual thread"
| mode                         | conc. |   reqs | wall ms |  req/s | p50 ms | p99 ms | peak platform threads |
errors |
| MVC Virtual threads          |   100 |   2000 |    2131 |    938 |    106 |    111 |                    33 |
0 |
| MVC Virtual threads          |  1000 |  10000 |    1259 |   7942 |    108 |    346 |                    33 |
0 |
CPU cores = 14 (Reactor Netty event-loop threads default to this number)
| mode                         | conc. |   reqs | wall ms |  req/s | p50 ms | p99 ms | peak platform threads |
errors |
| WebFlux non-blocking         |   100 |   2000 |    2185 |    915 |    107 |    122 |                    56 |
0 |
| WebFlux non-blocking         |  1000 |  10000 |    1241 |   8058 |    110 |    263 |                    56 |
0 |
| WebFlux + Thread.sleep (bad) |   100 |    500 |    5107 |     97 |    724 |   2479 |                    56 |
0 |
```

Reading the numbers

1. **At 100 clients all three are equal** (~930 req/s, p50 ≈ 105 ms): the ceiling is the clients (100 × 10 req/s). Below the thread limit platform threads are fine.
2. **At 1,000 clients platform threads hit the wall**: 200 workers × 10 req/s = 2,000 theoretical, 1,909 measured. Extra requests **queue**: median jumps from ~105 ms to **517 ms** (≈ 5 rounds of 100 ms).
3. **Virtual threads remove the wall without code changes**: 7,942 req/s at 108 ms median, ~4× platform throughput, with **33** platform threads in the JVM instead of 231 — each request's carrier served others while it slept.
4. **WebFlux matches virtual threads** (8,058 req/s, p50 110 ms; the difference is noise). Both are now CPU/load-generator bound. (p99 263–346 ms at 1,000 clients comes from client/server CPU contention in one JVM plus GC pauses — treat tail latencies here with care.)
5. **Blocking on an event loop is a disaster**: the same `Thread.sleep(100)` in a WebFlux handler gave **97 req/s**, p99 **2.5 s**. Each of 14 loops serves one sleeping request at a time (≤ 14 × 10 = 140 req/s; uneven connection spread lowers it), and every *other* request on a blocked loop waits too.

Which one should you choose?

Situation	Choose	Why
Typical CRUD/REST service on JDBC/JPA, blocking clients	**MVC + virtual threads**	same scalability for I/O-bound work, plain imperative code, all blocking libraries work, easy debugging
Streaming: SSE, WebSockets, long-lived connections, NDJSON, backpressure end-to-end	**WebFlux**	a stream is a `Flux`; backpressure and cancellation are built in
Gateways/proxies that mostly forward bytes (Spring Cloud Gateway's reactive variant)	**WebFlux**	minimal per-connection cost, non-blocking all the way
Heavy fan-out/composition: call 10 services, combine, time out, retry, race	either; **Reactor** operators are very expressive; structured concurrency (subtasks treated as one unit; still preview) is the imperative alternative	
CPU-bound work (image processing, crypto)	neither helps	you need more cores; keep it off event loops
Team unfamiliar with reactive, blocking dependencies you can't replace	**MVC + virtual threads**	a reactive stack with blocking calls inside is the worst of both

Virtual-thread caveats

- **Pinning**: before Java 24 a virtual thread blocking inside `synchronized` **pinned** its carrier (couldn't unmount). JEP 491 (JDK Enhancement Proposal; Java 24) fixed this for `synchronized`; native calls and some class-initialisation paths still pin. See Java ch15 §15.5.
- **Resources are still finite**: 10,000 virtual threads still share, say, 10 JDBC connections — the pool becomes the queue (Chapter 7). Limit downstream calls with semaphores (N holders at once) or bulkheads (capped resource pools per dependency).
- `ThreadLocal` **caches** (big per-thread buffers) multiply with millions of threads; prefer scoped values (immutable per-call alternative) or explicit passing.

In the interview

Q: WebFlux or virtual threads? "For I/O-bound request/response services I'd default to Spring MVC with `spring.threads.virtual.enabled=true`. In my measurements it matches WebFlux for a 100 ms I/O endpoint — about 8,000 req/s at 1,000 concurrent clients vs about 1,900 on 200 platform threads — and the code stays simple and debuggable. I'd pick WebFlux for streaming (SSE, WebSockets), gateways, or end-to-end backpressure and rich async composition, and only if the whole call path is non-blocking: a single blocking call on an event loop cut WebFlux to ~100 req/s."

- *Do virtual threads make code faster?* → No; they make *waiting* cheap, so more requests can wait at once. CPU-bound work gets no faster.
- *What is pinning?* → A virtual thread that can't unmount from its carrier while blocked (before Java 24 inside `synchronized`; still for native frames), so it holds a platform thread.

13.16 Common mistakes — checklist

Mistake	Symptom	Fix
Blocking call (JDBC, `RestClient`, `Thread.sleep`, `block()`) on an event loop	throughput collapses, timeouts under load; `block()` throws `IllegalStateException`	non-blocking alternative, or `fromCallable(..).subscribeOn(boundedElastic())`; BlockHound in tests
Pipeline built but not returned/subscribed	nothing happens (no save, no call)	return the `Mono`/`Flux`; chain with `then`/`flatMap`
`map` with an async function	`Flux<Mono<T>>`, inner calls never run	`flatMap` / `concatMap`
`Mono.just(expensiveCall())`	runs eagerly, even when not subscribed; retry doesn't repeat it	`Mono.fromCallable` / `Mono.defer`
Relying on `ThreadLocal` (MDC, security, tx)	values missing after a thread hop	Reactor `Context`, context-propagation
Unbounded `flatMap` / `onBackpressureBuffer()`	overwhelmed downstream, memory growth	concurrency argument, bounded buffers, `limitRate`
`@Table("book")` with Spring Data R2DBC on H2	`Table "book" not found (candidates are: "BOOK")`	omit the name or match the case
Missing `spring-boot-starter-webclient` in Boot 4	no `WebClient.Builder` bean	add the starter
Virtual-time test without `thenAwait`	test hangs	advance the clock; JUnit default timeout
Mixing MVC and WebFlux starters	Boot silently starts MVC	pick one, or set `spring.main.web-application-type`

13.17 In the interview

Q: What is reactive programming and what problem does it solve? "Programming with asynchronous data streams and a subscriber-controlled flow rate (backpressure). In a server no thread waits for I/O: a few event-loop threads handle thousands of concurrent requests because work resumes only when data arrives. Spring uses Project Reactor — `Mono` for 0..1, `Flux` for 0..N. Pipelines are lazy and cold until subscribed; in WebFlux the framework subscribes."

- *`Mono` vs `Flux`?* → 0..1 vs 0..N items, both ending with complete or error. *`flatMap` vs `concatMap`?* → Concurrent/interleaved vs sequential/ordered.
- *`publishOn` vs `subscribeOn`?* → `publishOn` moves the operators after it; `subscribeOn` moves the subscription (source) regardless of position.
- *Blocking library from WebFlux?* → `Mono.fromCallable(...).subscribeOn(Schedulers.boundedElastic())` — and question whether WebFlux fits.
- *Backpressure example?* → `request(n)` limits the publisher; `limitRate`, `onBackpressureDrop/Latest/Buffer` handle sources that can't slow down.
- *Testing?* → `StepVerifier` (virtual time for delays/retries), `WebTestClient`, `@WebFluxTest` / `@DataR2dbcTest` slices.
- *JPA with WebFlux?* → Not without blocking (wrap every call on `boundedElastic`). Use R2DBC, or MVC with virtual threads.
- *Reactive transactions?* → `ReactiveTransactionManager` (`R2dbcTransactionManager`); `@Transactional` on `Mono`/`Flux` methods or `TransactionalOperator`; the transaction lives in the Reactor `Context`, not a `ThreadLocal`.

13.18 Chapter TL;DR

- **Why**: thread-per-request wastes threads on waiting; a 200-thread Tomcat caps near 2,000 req/s for 100 ms I/O (measured 1,909). Event loops never wait. Same idea as Node's event loop.
- **Reactive Streams**: `Publisher` / `Subscriber` / `Subscription` / `Processor`; demand via `request(n)`; one terminal signal. JDK copy: `java.util.concurrent.Flow`.
- **Reactor**: `Mono` (0..1), `Flux` (0..N); lazy and cold until subscribed; `just` is eager, `defer` / `fromCallable` are lazy; don't `block()` in reactive code.
- **Operators**: `map`, `filter`, `flatMap` (concurrent, 6 s) vs `concatMap` (ordered, 12 s) vs `flatMapSequential`; `zip`, `concat`, `merge`, `switchIfEmpty`, `buffer`, `reduce`.
- **Errors** end the stream; `onErrorReturn/Resume/Map`, `retryWhen(Retry.backoff(3, 1s))` → attempts at 0/1/3/7 s, `timeout`.
- **Testing**: `StepVerifier`, virtual time (`withVirtualTime(() -> ...)` + `thenAwait`), JUnit default timeout against hangs.
- **Backpressure**: `request(n)`, `limitRate(10)` → requests 10 then 8s; overflow: DROP / LATEST / BUFFER(n) → error.
- **Schedulers**: `publishOn` (downstream) vs `subscribeOn` (source); `boundedElastic` for unavoidable blocking calls; never block event loops.
- **WebFlux**: annotated controllers returning `Mono` / `Flux`, functional `RouterFunction`s, SSE with `text/event-stream`; tests with `@WebFluxTest` + `WebTestClient` and `@SpringBootTest(RANDOM_PORT)` + `@AutoConfigureWebTestClient`. Boot 4: add `spring-boot-starter-webclient` for `WebClient`.
- **WebClient**: `retrieve`, `onStatus`, per-attempt `timeout`, filtered `retryWhen`; 5 × 300 ms calls in 322 ms with `flatMap`.
- **R2DBC**: non-blocking SQL, not an ORM; `ReactiveCrudRepository`, `DatabaseClient`, `TransactionalOperator`; `@DataR2dbcTest` doesn't roll back; mind identifier quoting.
- **Measured**: at 1,000 concurrent clients, MVC platform threads 1,909 req/s (p50 517 ms), MVC **virtual threads 7,942 req/s**, **WebFlux 8,058 req/s**, WebFlux with a blocking call **97 req/s**. Default to MVC + virtual threads for I/O-bound CRUD; choose WebFlux for streaming, gateways and end-to-end non-blocking pipelines.
- Next: Chapter 14 — Messaging, async and scheduling.